

Introduction to Vectors

↳ The heart of linear algebra

↳ Mainly two operations $\rightarrow$ Adding and Multiplying

Vectors and Linear Combination (1.1)

Addition: $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ and $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ add to $v+w = \begin{bmatrix} v_1+w_1 \\ v_2+w_2 \end{bmatrix}$

Multiplication: $2v = \begin{bmatrix} 2v_1 \\ 2v_2 \end{bmatrix} = v+v$
 $-v = \begin{bmatrix} -v_1 \\ -v_2 \end{bmatrix}$

⊗ Vector in three dimensions works the same way.

$v = \begin{bmatrix} u \\ x \\ y \end{bmatrix}$ is also written as $v = (u, x, y)$

⊗ If u, v and w are three vectors in three dimensional space. -

↳ The combinations cu fill a line through $(0,0,0)$.

↳ The combinations $cu+dv$ fill a plane through $(0,0,0)$.

↳ The combinations $cu+dv+ew$ fill three-dimensional space.

Problem Set 1.1

① a) $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$

↳ Here, $v_2 = 3v_1 \rightarrow$ They are linearly dependent and occupying only 1D space.

↳ So geometrically, this is a line of $\mathbb{R}^3$.

b) $v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$

↳ Here, v_1 and $v_2 \rightarrow$ They are linearly independent, occupying only 2D space by creating two different lines.

↳ So geometrically, this is a plane of $\mathbb{R}^3$.

c) $v_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$ and $v_3 = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$

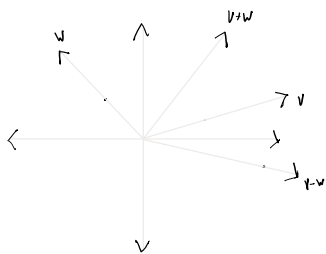
↳ Here v_1, v_2 and $v_3 \rightarrow$ all of them are linearly independent to each other, occupies 3D space by creating three different.

② $v = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$



② $v = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$

$v+w = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $v-w = \begin{bmatrix} 6 \\ -1 \end{bmatrix}$



③ $\Rightarrow$ Easy stuff

④ $v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$\therefore 3v+w = 3\begin{bmatrix} 2 \\ 1 \end{bmatrix} + 1\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$

$\therefore cv+dw = c\begin{bmatrix} 2 \\ 1 \end{bmatrix} + d\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2c+d \\ c+2d \end{bmatrix}$

⑤ $u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}$, $w = \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix}$

$\hookrightarrow u+v+w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow u+v = -w$

and, $2u+2v+w = \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}$

Since, $u+v = -w$, they are linearly dependant. So, they are in 2D. They lie in a plane.

$\Rightarrow 1-2+1=0, 0+1-1=0$

⑥ $v = (1, -2, 1)$
 $w = (0, 1, -1)$ both of these components add to 0

Now, $c\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + d\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} c \\ -2c+d \\ c-d \end{bmatrix}$

if, $\begin{bmatrix} c \\ -2c+d \\ c-d \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ -6 \end{bmatrix} \rightarrow \begin{cases} c \text{ must be } 3 \\ -2c+d=3 \\ c-d=-6 \end{cases} \begin{cases} c=3 \\ d=9 \end{cases}$

But if, $\begin{bmatrix} c \\ -2c+d \\ c-d \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ -6 \end{bmatrix} \rightarrow \begin{cases} c=3 \\ -2c+d=3 \\ c-d=-6 \end{cases} \begin{cases} c=-9 \\ d=-15 \end{cases}$

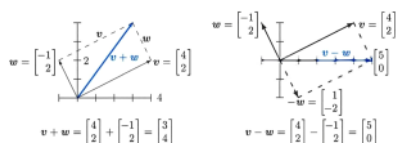
c can't be 3 and -9 at the same time.

So there is no solution.

⑦ $\Rightarrow$ Just graph pointing

⑧

Figure 1.1 $\rightarrow$



The other diagonal.
Or it can also be $w - v$.

$\rightarrow$ Adding them up will give you $2v$ or $2w$

⑨ (Irrelevant)

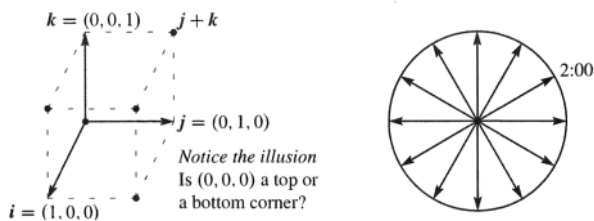
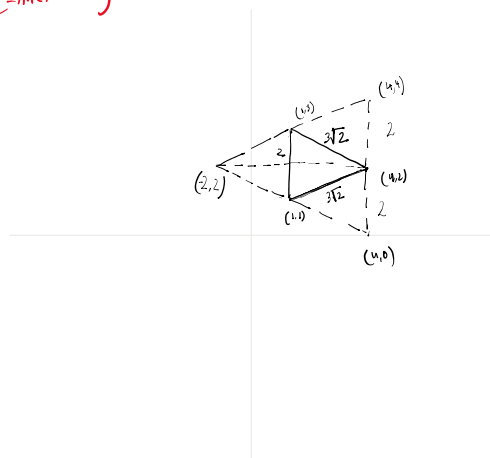


Figure 1.4: Unit cube from i, j, k and twelve clock vectors.

⑩

$$i = (1, 0, 0)$$

$$i + j = (1, 1, 0)$$

$$j = (0, 1, 0)$$

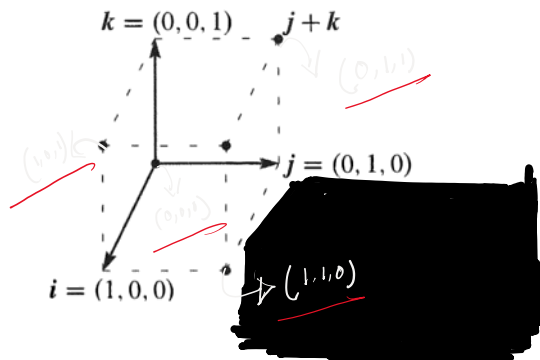
$\hookrightarrow$ Represents a point on the edge of the cube in the xy plane, at the intersection of $x=1$ and $y=1$ edges.

cube in the xy plane, at the intersection of $x=1$ and $y=1$ edges.

Now, vector sum of $i = (1, 0, 0)$, $j = (0, 1, 0)$ and $k = (0, 0, 1)$ is

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \rightarrow (1, 1, 1) \rightarrow \text{the opposite corner of origin.}$$

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(You can find the center points by yourself)

12 $i = (1, 0, 0)$

$$i + j = (1, 1, 0)$$

Their linear combination can be written as:

$$v = c(1, 0, 0) + d(1, 1, 0) = (c+d, d, 0)$$

So, the general point for any point on the plane is $(c+d, d, 0)$

$$\text{Here, } (x, y, z) = (c+d, d, 0) \Rightarrow \left. \begin{array}{l} x = c+d \\ y = d \\ z = 0 \end{array} \right\} x = c+y.$$

↳ This consists of all points that satisfy the equation $x = c+y$, where c is a constant which describes a family of lines parallel to $x=y$ in the xy plane.

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(a) $\text{sum } v = 0$

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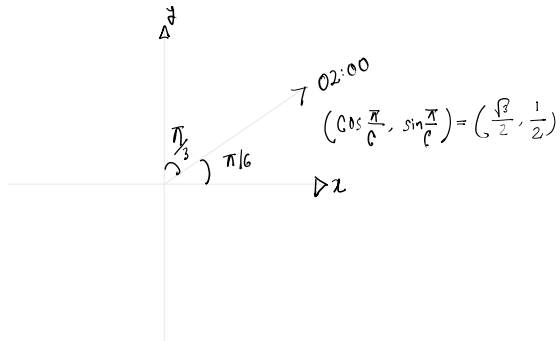
a) $\sum \vec{v} = 0$

the xy plane.

$\hookrightarrow$ Every vector has their opposite ends.

b) If 2:00 is removed, then the only remaining vector is 8:00.

c)



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